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SEAT No. :

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P1391

BE/Insem./APR-21

B.E. (Mechanical)

INDUSTRIAL ENGINEERING

(2012 Course) (Semester - II) (Elective - III) (402049 C)

Time : 1½ Hours]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q1 or 2, Q3 or 4, Q5 or 6.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) Describe briefly the contribution of F W Taylor in to the management field. [5]

b) Explain a line and staff organization. What are merits and demerits? [5]

OR

Q2) a) What is productivity? Explain relation between work-study and productivity. [5]

b) What are the key functions of management? Discuss them briefly. [5]

Q3) a) Describe in brief the steps involved in carrying out a method study. [5]

b) What are Therbligs? When they are used? List five therbligs with symbols. [5]

OR

Q4) a) Describe with suitable example symbols used in Method study. [5]

b) What are the principles of motion economy? Explain. [5]

Q5) a) Define time study and explain its objectives. [4]

b) A time study was conducted on a job consists of three elements. Stop watch reading in hundredth of a minute is given. Using cumulative timing method along with rating factors, Calculate the standard time if allowance is 12%. [6]

P.T.O.

Element	Stopwatch readings					Rating
	1	2	3	4	5	
A	10	73	139	203	266	80
B	25	88	155	218	280	100
C	64	128	193	257	320	110

OR

- Q6) a)** A worker operating on a machine performs the following operations. the description of element, their observed time and ratings are given calculate the standard time for the component. [6]

Element	Observed time	Rating %	Relaxation Allowance %
A	0.25	80	10
B	0.09	100	11
C	2.8	90	12
D	0.05	80	10
E	.15	110	11

- b) What is work sampling? What are its merits and limitations? Explain [4]

